

DevFusion 4.0 : The Developers Hackathon



RULES HERE 

Choose any 1 Problem statement and start doing.

Problem Statement 2: AI-Powered Recruitment & Applicant Tracking System (ATS)

Difficulty

Advanced

Domain

HRTech • AI • SaaS • Enterprise

Duration

24–48 Hours

Background

Recruitment today involves managing hundreds of applications, manually screening resumes, scheduling interviews, tracking candidate progress, collecting interviewer feedback, and communicating with applicants across multiple platforms. This process is often slow, repetitive, and inefficient.

Build a production-ready AI-powered Applicant Tracking System (ATS) that enables companies to manage the entire hiring lifecycle—from posting jobs to onboarding selected candidates—with secure authentication, role-based access, AI-assisted resume analysis, interview scheduling, coding assessments, and analytics.

Objective

Develop a full-stack web application that allows recruiters, hiring managers, interviewers, and candidates to collaborate on the recruitment process through a centralized platform.



The application should support:

- Job Management
 - Candidate Management
 - Resume Upload & Parsing
 - AI Resume Analysis
 - Interview Scheduling
 - Coding Assessments
 - Email Notifications
 - Analytics Dashboard
 - Offer Letter Generation
 - Candidate Portal
-

Mandatory Tech Stack

Frontend

- Next.js / React
- TypeScript
- Tailwind CSS
- Responsive Design

Backend

- Node.js
- Express / Next.js API
- MongoDB/PostgreSQL

Authentication

- Google OAuth
- Email & Password
- JWT / Secure Sessions

Deployment

- Vercel
 - Railway / Render
 - Docker (Bonus)
-

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Authentication Requirements

Sign Up

Support

- Google Login
 - Email Registration
-

Login

Support

- Google OAuth
 - Email Login
-

Email Verification

New users must verify their email before accessing recruiter features.

Forgot Password

Implement

- OTP Verification
OR
 - Password Reset Link
-

Session Security

- Secure Cookies
 - JWT Refresh Tokens
 - Device Session Management
-

Two-Factor Authentication (Bonus)

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Recruiters and Admins can enable OTP-based 2FA.

User Roles

Candidate

Can

- Create Profile
- Upload Resume
- Apply for Jobs
- Track Applications
- Schedule Interviews
- Take Coding Tests
- Receive Offer Letters
- Accept/Reject Offers

Cannot

- Access recruiter data
-

Recruiter

Can

- Post Jobs
- View Applications
- Shortlist Candidates
- Schedule Interviews
- Send Emails
- Generate Offer Letters

Cannot

- Change company settings
-



Hiring Manager

Can

- Review Shortlisted Candidates
 - Approve Hiring Decisions
 - Give Feedback
 - View Analytics
-

Interviewer

Can

- View Assigned Interviews
- Submit Interview Feedback
- Score Candidates

Cannot

- View salary information
-

Admin

Full Control

Can manage

- Users
 - Roles
 - Companies
 - Jobs
 - Assessments
 - Analytics
 - Platform Settings
 - Audit Logs
-

Landing Page

Include

- Hero Section
 - Product Features
 - Customer Testimonials
 - Pricing
 - FAQ
 - Contact
 - Footer
 - Responsive Navigation
 - Dark Mode
 - SEO Optimization
-

Recruiter Dashboard

Widgets

- Total Jobs
 - Active Candidates
 - Today's Interviews
 - Pending Reviews
 - Offer Acceptance Rate
 - Hiring Funnel
 - Candidate Conversion Rate
 - Monthly Hiring Chart
 - Recent Activity
-

Candidate Dashboard

Display

- Profile Completion
- Applied Jobs
- Interview Schedule
- Coding Assessments
- Offer Letters
- Notifications



- Resume Score
 - AI Suggestions
-

Job Management Module

Recruiters can

Create

- Job Title
- Department
- Location
- Salary Range
- Experience Required
- Skills Required
- Employment Type
- Work Mode
- Deadline
- Job Description
- Company Details

Manage

- Edit
 - Close
 - Duplicate
 - Delete Jobs
-

Candidate Profile

Fields

- Profile Picture
- Name
- Email
- Phone
- Location
- Education



- Experience
 - Skills
 - Certifications
 - Portfolio
 - GitHub
 - LinkedIn
 - Resume Upload
 - Cover Letter
-

Resume Upload

Support

- PDF
- DOCX

Maximum size

10 MB

Validate

- File Type
 - Virus Check (Bonus)
 - Duplicate Upload
-

AI Resume Parsing

Extract automatically

- Name
- Email
- Phone
- Skills
- Education
- Experience
- Projects
- Certifications

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- Languages
- Total Experience

Auto-fill candidate profile.

AI Resume Matching

Generate a matching score between candidate resume and job description.

Display

- Match Percentage
- Missing Skills
- Strengths
- Weak Areas
- Recommendations

Example

- Overall Match: 87%
-
- Strong Skills
-
- ✓ React
- ✓ Node.js
- ✓ MongoDB
-
- Missing Skills
-
- ✗ AWS
- ✗ Docker
-
- Recommendation
-

Good fit for interview.

Job Search

Candidates can filter by

- Location
 - Salary
 - Skills
 - Company
 - Experience
 - Remote
 - Hybrid
 - Full Time
 - Internship
-

Application Workflow

Stages

Applied



Resume Screening



Shortlisted



Technical Interview



HR Interview



Offer



Hired

OR

Rejected

Recruiters can drag-and-drop candidates between stages.

Interview Scheduler

Recruiters can

- Select interviewer
- Choose date/time
- Generate meeting link (Zoom/Google Meet)
- Send invitations

Candidates receive

- Email
 - Dashboard Notification
 - Calendar Reminder
-

Coding Assessment Module

Recruiters can create tests with

- MCQs
- Coding Problems
- SQL Queries
- Debugging Tasks

Features

- Countdown Timer
 - Auto Submission
 - Tab Switch Detection
 - Code Editor
 - Test Analytics
 - Plagiarism Detection (Bonus)
-



Interview Feedback

Interviewers submit

- Technical Skills
- Communication
- Problem Solving
- Teamwork
- Leadership
- Overall Rating
- Comments

Hiring managers can compare interviewer feedback before making a decision.

Offer Letter Generator

Recruiters can generate offer letters using templates.

Fields

- Candidate Name
- Role
- Salary
- Joining Date
- Location
- Benefits

Candidates can

- Download PDF
 - Accept Offer
 - Reject Offer
-

Email System

Automatic emails for

- Application Confirmation



- Shortlisting
 - Interview Invitation
 - Assessment Link
 - Offer Letter
 - Rejection
 - Joining Instructions
-

Notifications

Real-time notifications for

- New Applications
 - Interview Updates
 - Assessment Results
 - Offer Status
 - Profile Completion
-

Search

Global Search

Search

- Candidates
 - Jobs
 - Companies
 - Recruiters
 - Interviews
-

Reports & Analytics

Recruiter Dashboard should include

- Applications Per Job
- Hiring Funnel
- Time-to-Hire



- Offer Acceptance Rate
- Candidate Source Analysis
- Diversity Metrics (optional)
- Recruiter Performance
- Interview Success Rate

Visualize data using charts and tables.

Company Profile

Fields

- Logo
 - Name
 - Website
 - Industry
 - Company Size
 - Description
 - Social Links
 - Office Locations
-

Admin Panel

Manage

- Users
 - Companies
 - Jobs
 - Assessments
 - Recruiters
 - Permissions
 - Platform Settings
 - Audit Logs
 - Reports
-

UI/UX Requirements

Must include

- Responsive Layout
 - Dark & Light Mode
 - Modern Dashboard
 - Kanban Board for Applications
 - Skeleton Loaders
 - Empty States
 - Toast Notifications
 - Smooth Animations
 - Keyboard Accessibility
 - Mobile-Friendly Design
-

Security Requirements

- Password hashing (bcrypt/Argon2)
 - RBAC (Role-Based Access Control)
 - Input validation
 - Rate limiting
 - Secure cookies
 - XSS protection
 - CSRF protection (where applicable)
 - File upload validation
 - API authorization middleware
 - Audit logging
 - Secure environment variables
-

Database Entities

Minimum collections/tables

- Users
- Roles
- Companies
- Jobs



- Candidates
 - Applications
 - Resumes
 - Resume Analyses
 - Interviews
 - Interviewers
 - Feedback
 - Coding Assessments
 - Assessment Attempts
 - Offer Letters
 - Notifications
 - Activity Logs
 - Settings
-

Bonus Features

- AI chatbot for candidate FAQs
 - AI-generated interview questions based on job descriptions
 - AI-powered cover letter generation
 - AI interview feedback summarization
 - Live collaborative interview notes
 - Resume version history
 - Candidate referral system
 - Bulk resume import
 - Public careers page
 - Calendar integration (Google/Microsoft)
 - WebSocket-based live application updates
 - PWA support
 - Multi-language support
 - Dockerized deployment
 - CI/CD pipeline
 - API documentation (Swagger/OpenAPI)
-



Judging Criteria

Criteria	Weight
Core Functionality	30%
AI Features & Innovation	15%
UI/UX Design	15%
Code Quality & Architecture	15%
Authentication & Security	10%
Database Design	5%
Performance & Scalability	5%
Deployment & Documentation	5%

Expected Deliverables

- GitHub repository with clean commit history
- Live deployed application
- README with setup and architecture details
- API documentation (Swagger/OpenAPI)
- Database schema/ER diagram
- Sample recruiter and candidate credentials



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- Environment variable template (.env.example)
- Demo video (3-5 minutes)
- Postman collection or API export
- CI/CD configuration (bonus)
- Docker setup (bonus)
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